

Unit

3

Lesson

1



Beyond Imagination.

Smart Sensors in Action.

Time Frame: [90 min].

Ready, Set, Learn!

You Need to Know

- Build a smart trash bin that signals when it's full.
- Combine coding and hardware to bring your design to life.
- Develop problem-solving skills by turning an ordinary bin into a clever, automated tool.

Challenge questions of the lesson.

1. How does the smart bin "know" when trash is near or when it's full? Where do you think the sensors need to be placed so they can detect things accurately?
2. Why is it helpful for the smart bin to have automatic lights (LED) and be able to sort trash by itself? How does this make managing trash better for people and the environment?
3. If you could add one new smart feature to our bin, or make its current features even better, what would it be? For example, could it use another kind of sensor, or do something else special?

SEL Relation

- **Creative Expression (Self-Awareness):** Students will express their understanding of the smart bin's automated functions and the logic from the coding game, potentially by sharing ideas for new features or explaining how the existing features make the bin "smart."
- **Social Awareness (Environmental & Community Impact):** Students will understand how the smart bin's features contribute to a cleaner environment and more efficient waste management in the community, fostering a sense of responsibility and connection to the "Turning Trash into Power" concept.
- **Problem-Solving and Perseverance (Self-Management):** Students will practice problem-solving skills when engaging with challenges in the coding game (e.g., figuring out the correct sequence of code blocks) and understanding how the smart bin itself solves real-world waste problems.
- **Communication and Collaboration (Relationship Skills):** If working in pairs or groups for the coding game or reflection activities, students will need to communicate their coding strategies, explain their understanding of the smart bin's logic, and build on each other's ideas.

Engage

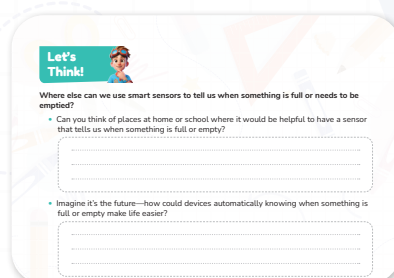
Teacher Role:

- **Facilitator:** Guide the discussion about the bin's overflow problem, highlighting the students' creative ideas for indicating fullness.
- **Questioner:** Pose questions to encourage students to think about how they could make the bin clearly show when it's full, building on the IR sensor idea.
- **Connector:** Help students connect the idea of an indicator light or other signals to things they already know and understand (battery low light on a phone, car dashboard warnings, traffic lights).
- **Enthusiast:** Show excitement about the students' imaginative ideas for indicators to foster a positive and creative learning environment.
- **Guide:** Steer the conversation towards the practical aspects of making the full-bin indicator work, hinting at the need to understand how LEDs or other visual/audible alerts function.

Expected From Students:

- **Active Listening:** Students will listen attentively to the problem (bin overflow) and engage with the dialogue proposing solutions.
- **Creative Thinking:** Students will be inspired to think creatively about how to make a smart bin effectively signal when it's full.
- **Idea Sharing:** Students will share their own ideas about how to use lights, sounds, or other visual cues to indicate bin fullness.
- **Initial Problem-Solving:** Students will begin to think about the basic requirements for making an indicator work with the IR sensor.
- **Curiosity:** Students will become curious about how LEDs or other warning indicators work and how they can be connected to an IR sensor.

Parts of the books included.



Story



1. **Set the Scene with Enthusiasm:** Begin by saying: "Class, today we're going to solve a real-world problem and make something amazing a smart bin that tells us when it's full, so no more overflowing trash!" Then ask: "Why do you think having a clear way to know when the trash bin is full might be helpful or important for our school community?"
2. **Hook with the Story Dialogue:** Introduce the scenario: "Let's listen to Mr. Yusef, Adam, Laila, and one of our classmates as they discuss this very problem!" Read the story aloud using different voices for each character to bring energy and excitement:

Possible Strategies for Implementation:

- Strategy 1: Role Play the Dialogue.
 - How to Implement: Let students take on the parts of Mr. Yusef, Adam, Laila, and Student1. This builds engagement and helps them connect with the problem and the proposed solution.
 - Why it Helps: Encourages active listening and empathy for the real-world problem, making the lesson more relatable.
- Strategy 2: Brainstorm Indicator Ideas.
 - How to Implement: After reading the dialogue, ask: "What do we already know about things that show status or give alerts?" "Where do we see indicator lights or sounds in real life that tell us something important (like a battery low light, a traffic light, a warning buzzer)?" Write their responses on the board.
 - Why it Helps: Activates prior knowledge and broadens their thinking beyond just lights to other forms of clear communication.
- Strategy 3: Visual Thinking Strategy: Real-World Indicators.
 - How to Implement: Show real LEDs, simple indicator lights (from an old appliance), or videos of warning systems (like a "full" light on a printer or a car's fuel light). Let students observe and describe what they see, how these indicators work, and how they might use similar concepts in a smart bin design.
 - Why it Helps: Provides concrete examples and sparks ideas for how to clearly signal bin fullness.
- Strategy 4: Use Think-Pair-Share: Design Our Bin's "Full" Signal.
 - How to Implement: Ask students: "If you could design the perfect way for our smart bin to tell us it's full, what would it look like? Would it be a light? What color? Would it make a sound? Where would it be?" Let them first think individually, then share with a partner, and finally discuss with the class.
 - Why it Helps: Encourages individual creativity and collaborative problem-solving for the indicator mechanism.
- Strategy 5: Connect to Real-World Purpose: Solving a Community Problem.

- How to Implement: Say: "These indicators aren't just for fun they are crucial for solving a real problem! A clear 'full' signal can help Mr. Yusef manage the trash more efficiently, prevent overflows, keep our school cleaner, and contribute to a healthier environment for everyone!"
- Why it Helps: Reinforces the practical application and social impact of their engineering solution.

Let's Think!



Objective:

Students will extend their understanding of sensor systems by brainstorming and identifying diverse real-world scenarios where knowing if something is "full" or "empty" is crucial within home, school, and future contexts. This activity fosters creative thinking, problem-solving skills, and helps students connect their smart bin project to broader applications of technology.

Possible Answers

1. Think about your home or school:

- Trash Cans (like our smart bin!): To know when they are full and need emptying.
- Pet Food/Water Bowls: To alert when they are empty and need refilling.
- Soap Dispensers: To indicate when hand soap is low or empty.
- Paper Towel/Toilet Paper Holders: To signal when rolls are running out.
- Laundry Detergent Dispenser: To alert when it's low.
- Water Cooler/Dispenser: To know when the water bottle is empty.
- Recycling Bins: To signal when they are full and ready for collection.
- Pencil Sharpeners: To alert when the shaving compartment is full.
- Classroom Supply Bins: To know when a bin of markers, paper, or craft supplies is getting low.

2. Imagine you're in the future:

- Smart Pantries/Refrigerators: To automatically reorder groceries when items are low or empty.
- Self-Watering Plant Systems: To know when the water reservoir is empty and needs refilling.
- Robot Vacuum Cleaners: To alert when the dustbin is full and needs emptying.
- Smart Coffee Makers: To know when the water reservoir or coffee bean hopper is empty.
- Medicine Dispensers: To alert when a pill compartment is empty.
- Automated Composting Bins: To know when they are full and ready for emptying or processing.

Possible Strategies for Implementation:

Strategy 1: Think-Pair-Share (Focus on Real-World Applications).

- How to Implement:
 - Think: Ask students to individually think about the main question: "Where else can we use smart sensors to know if something is full or needs to be emptied?" Prompt them with the specific sub-questions: "Think about your home or school," and "Imagine the future." Give them a few minutes to jot down their initial ideas.
 - Pair: Have students turn to a partner and share their ideas. Encourage them to listen to each other and add new ideas that their partner thought of.
 - Share: Bring the class back together and ask pairs to share their most interesting or surprising ideas with the whole group.
- Benefit: This strategy encourages individual reflection, peer collaboration, and broadens the scope of ideas shared with the class.

Strategy 2: Brainstorm Web / Mind Map: "Full/Empty Sensor Ideas"

- How to Implement: Start a large web or mind map on the board (or use a digital tool). Write "Full/Empty Sensor Ideas" in the center. As students share their ideas (from the Think-Pair-Share or new ones), write them around the center, categorized by home, school, or future concepts. Draw lines connecting related ideas.
- Benefit: Provides a visual representation of all the diverse applications, making connections clear and showing the wide impact of sensor technology.

Strategy 3: "Problem-Solution Scenario Cards" (Optional Extension)

- How to Implement: Prepare simple cards with "full/empty" problems ("The dog's water bowl is empty," "The printer ink is low," "The mailbox is full"). Students pick a card and briefly describe how a sensor system could solve that problem, mentioning what kind of sensor might be needed.
- Benefit: Challenges students to apply their knowledge to specific hypothetical situations, reinforcing their understanding of sensor functionality and practical solutions
- You can use this link: <https://gemini.google.com/share/e60a3536a04b>



Explorer's Toolkit



Teacher Role:

- Introducer: Introduce the activity as an exploration of building a "smart helper" circuit with two special "eyes" – IR sensors. Explain that one sensor will watch the wet trash side, and the other will watch the dry trash side of their bin. Emphasize that these "eyes" will automatically tell them if each part of the bin is full, helping them know exactly when to empty it.
- Safety Guide: Emphasize the importance of handling electronic components carefully and ensuring the Arduino is unplugged before making or changing any connections.

- **Facilitator:** Guide students through the process of identifying and connecting the components needed for the dual-IR sensor circuit (two IR sensors, Arduino, breadboard, wires, and perhaps indicator LEDs for each side).
- **Observer:** Observe students as they build their circuits and test them, noting their ability to follow instructions, connect multiple sensors, and understand how each sensor detects fullness in its respective bin compartment.
- **Questioner:** Encourage students to describe what they observe when trash is placed near each IR sensor and to think about how having two "eyes" is more helpful than just one for their dual-compartment bin.

Expected From Students:

- **Hands-on Participation:** Students will actively participate in building the dual-IR sensor circuit on the breadboard with the Arduino.
- **Following Instructions:** Students will follow the instructions to connect the two IR sensors and any associated indicator LEDs correctly.
- **Observation:** Students will carefully observe what happens when they place an object (simulating trash) near each IR sensor, noting how the system responds.
- **Understanding Dual Detection:** Students will begin to understand how having separate sensors for different compartments allows for precise "full" detection for both wet and dry waste sides of the bin.

Parts of the books included.



DIY Lab

You're going to connect a smart circuit that helps our trash bin do its job better! It includes two special "eyes"—IR sensors:

- One sensor keeps an eye on the wet waste section.
- The other watches the dry waste section.

These sensors can automatically detect when each side of the bin is full.

That way, we'll always know the right time to empty it!

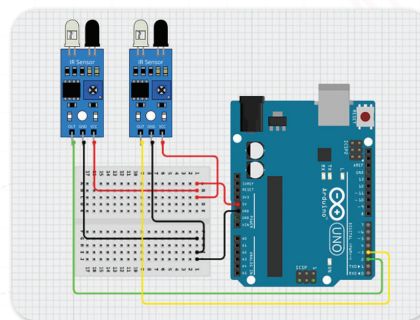


Figure 28: Dual IR Sensor Circuit with Arduino.

STEAM SCOUTS

47



DIY Lab

Objective:

Students will construct a functional electronic circuit using an Arduino, two IR Infrared Sensor Modules (acting as "eyes" for wet and dry trash compartments), and indicator LEDs. This activity helps students explore how multiple sensors can be integrated to detect conditions in different areas, providing automatic alerts for "full" or "empty" states in a

dual-compartment smart bin through hands-on investigation.

Materials Needed:

- Arduino Uno Board
- Breadboard
- 2 x IR Infrared Sensor Module.
- Jumper Wires (Assorted lengths, male-to-male).
- USB Cable (for connecting Arduino to computer).

Safety First! Remind students to handle all components carefully and ensure power is off (Arduino unplugged) before making connections.

Step-by-Step Instructions:

4. Wiring the Components:

- Connect the VCC pin of both IR sensors to 5V on the Arduino.
- Connect the GND pin of both IR sensors to GND on the Arduino.
- Connect the OUT pin of the first sensor to digital pin 2 on the Arduino.
- Connect the OUT pin of the second sensor to digital pin 3 on the Arduino.

Uploading the Code: :(don't let them write the whole code, prepare another version that includes guide lines).

- Open the Arduino IDE and write the following code:

```
int ledPin = 2;
int sensorPin = 7;
int sensorValue = 0;
void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(sensorPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  sensorValue = digitalRead(sensorPin);
  if (sensorValue == HIGH) {
    digitalWrite(ledPin, HIGH);
    Serial.println("Bin is FULL!");
  } else {
    digitalWrite(ledPin, LOW);
  }
  delay(500); // Optional: helps avoid rapid flickering
               // or message spam
}
```

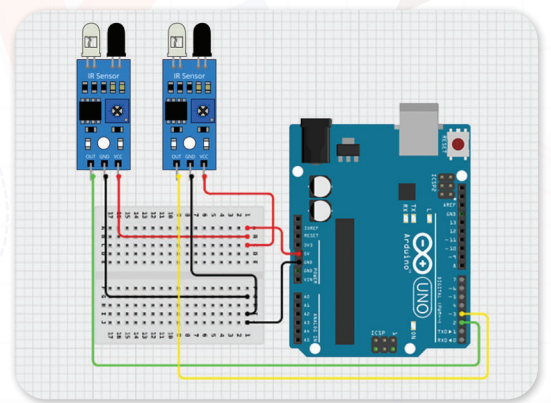


Figure 28: Dual IR Sensor Circuit with Arduino.

5. Running the Experiment:

- Upload the code to the Arduino board.
- Open the Serial Monitor in the Arduino IDE (Tools ▢ Serial Monitor).
- Observe the sensor values (0 or 1) as objects pass in front of the IR sensors.

Possible Strategies for Implementation:

Strategy 1: Model First: The Dual Detection Demo

- How to Implement: Demonstrate how to connect both IR sensors to the Arduino, along with their respective indicator LEDs. Show how when an object is placed near IR Sensor 1, LED 1 lights up, and when near IR Sensor 2, LED 2 lights up. Talk through the purpose of each sensor for its specific bin compartment.
- Benefit: Provides a clear visual example of the completed circuit and its functionality, specifically highlighting the role of each sensor.

Strategy 2: Pair Collaboration: Two Sets of "Eyes"

- How to Implement: Let students work in pairs. One student can focus on connecting the "wet side" sensor and its LED, while the other connects the "dry side" sensor and its LED. They can then collaboratively test both sides.
- Benefit: Encourages teamwork, shared responsibility, and helps students understand the separate but complementary roles of each sensor.

Strategy 3: Visual Checklists with Sensor Labels

- How to Implement: Use visual aids or icons next to the steps and components, explicitly labeling them "IR Sensor (Wet)" and "IR Sensor (Dry)" or "LED (Wet)" and "LED (Dry)". This supports emerging readers or ELL students and reinforces the dual-compartment concept.
- Benefit: Clarifies the specific role of each component in the dual-sensor setup, reducing confusion.

Strategy 4: Compare & Contrast Discussion: Why Two Eyes?

- How to Implement: Facilitate a group discussion where students compare the behavior and purpose of the two IR sensors. Use guiding questions like:
 - "What's the difference in what IR Sensor 1 is watching compared to IR Sensor 2?"
 - "Why is it more helpful to have two IR sensors for our smart bin instead of just one?"
 - "How does each sensor 'tell' us that its side is full?"
- Benefit: Promotes critical thinking about the design choices and the advantages of a multi-sensor system for a dual-compartment bin.

Explain



Teacher Role

- **Introducer:** Introduce the activity as an exploration of how innovative entrepreneurs are turning waste into valuable resources and products. Explain that they will be watching a video showcasing different examples of this "trash-to-treasure" approach.
- **Observation Guide:** Direct students' attention to the various examples presented in the video, such as turning plastic into bricks, coconut shells into cooking fuel, and glass into sand. Encourage them to note the specific waste materials being used and the resulting products.
- **Discussion Facilitator:** Lead a post-viewing discussion to explore the different initiatives, their environmental impact, and the entrepreneurial spirit behind them.
- **Connection Maker:** Help students connect these examples to the broader concept of sustainability and the importance of finding creative solutions to the global waste problem.

Expected From Students:

- **Attentive Viewing:** Students will watch the video carefully to observe the different entrepreneurs and their innovative approaches to waste management.
- **Feature Identification:** Students will identify and note the different types of waste materials being used and the products being created from them.
- **Critical Analysis:** Students will begin to analyze the environmental and economic benefits of these waste-to-energy initiatives.
- **Active Participation:** Students will share their observations and ideas during the post-viewing discussion.
- **Parts of the books included.**



Screen Lessons

Ready to turn trash into treasure? Discover how waste can power our world in this amazing video!

https://www.youtube.com/watch?v=QxjEGG5JVqg&ab_channel=SCIENCE%26FUN



Screen Lessons

Objective:

Students will explore how waste can be transformed into valuable resources and products by observing real-world examples of entrepreneurial ventures. By watching the video "Meet 8 Young Founders Turning Trash into Cash", students will understand how innovative thinking and engineering can contribute to a more sustainable future by turning waste into energy and other useful materials.

Video link: https://www.youtube.com/watch?v=QxjEGG5JVqg&ab_channel=SCIENCE%26FUN



Possible Strategies for implementation:

1. Before Watching (Set the Purpose):

- Prompt Curiosity with a Question: Ask: "What if we could turn our trash into something valuable instead of just throwing it away? What are some things we could make from waste?" Let students share their initial ideas.
- Activate Background Knowledge: Briefly review the concept of recycling and its importance. Then, introduce the idea of "upcycling" and turning waste into something even more valuable.
- Introduce a Focus: Tell students to look for the different types of waste being used, the products being made, and the impact of these initiatives while watching the video.

2. During Watching (Active Engagement):

- Visual Cues & Note-Taking: Ask students to jot down the different types of waste and the products being made from them as they watch.
- Pause & Discuss: Pause at key moments to ask guiding questions like:
 - "What kind of waste is this entrepreneur using?"
 - "What are they making from this waste?"
 - "How does this help the environment?"
 - "Could we do something similar in our community?"

3. After Watching (Reflection & Application):

- Think-Pair-Share: Students reflect individually, then discuss with a partner:
 - "What was the most surprising thing you learned about turning trash into treasure?"
 - "Which of these ideas was the most interesting or impactful, and why?"
 - "What are some challenges that entrepreneurs might face?"
- Creative Design Challenge Link: Prompt students to think about a specific waste stream (plastic bottles, food scraps) and brainstorm a product or process to turn it into something useful.
- Connection to Career Readiness: Discuss the types of jobs and skills involved in waste management, recycling, and sustainable innovati.

Elaborate and Evaluate



Teacher Role:

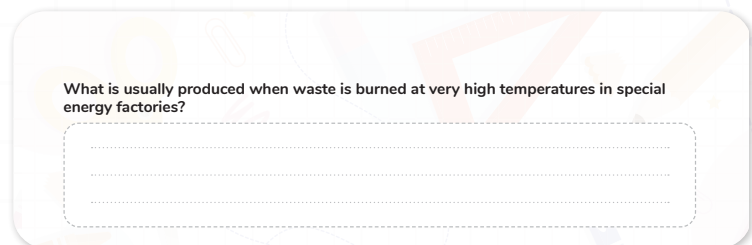
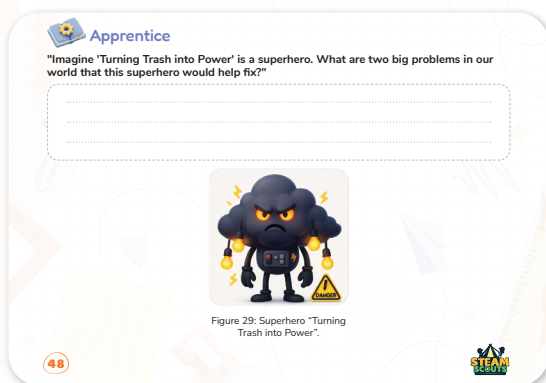
- Facilitator: Guide students through the conceptual discussion about "Turning Trash into Power" as a superhero fighting real-world problems.

- Clarifier: Explain the process of converting waste into energy (what happens when waste burns to make heat).
- Observer: Monitor student engagement during discussions on environmental problems and solutions.
- Encourager: Reinforce creative thinking about global environmental challenges and their solutions. Provide positive feedback for correct vocabulary usage when describing waste-to-energy concepts.

Expected from Students:

- Conceptual Understander: Identify and describe two major "villains" (problems) that "Turning Trash into Power" would fight and solve in our world (e.g., pollution, overflowing landfills, energy shortages).
- Process Explainer: Explain what heat from burning waste mostly makes in power factories (e.g., steam to turn turbines for electricity).
- Creative Thinker: Demonstrate creative thinking in articulating environmental problems and solutions.
- Communicator: Clearly describe the problems that "Turning Trash into Power" addresses and explain how the heat from burning waste is used to create power.

Parts of the books included.



Apprentice

Objective:

Students will be able to identify key environmental problems that "Turning Trash into Power" can help solve, and describe the basic process of how heat from burning waste is used to create power. This supports foundational understanding of sustainable energy solutions and builds vocabulary related to waste management and energy generation.

Answers:

1. If "Turning Trash into Power" was a superhero, which two "villains" (problems) would it fight and solve in our world?
 - Villain 1: Overflowing Landfills / Waste Pollution: Turning trash into power reduces the amount of garbage that goes into landfills, saving space and preventing pollution of land and water.

- Villain 2: Energy Shortages / Need for Fossil Fuels: Waste-to-power provides a new source of energy, reducing our reliance on burning fossil fuels like coal, oil, or gas, which helps fight climate change.

2. When waste burns and gets super-hot in special power factories, what does the heat mostly make?

- The heat mostly makes steam. This super-hot steam then spins giant machines called turbines, which are connected to generators that produce electricity.

Possible Strategies for Implementation:

Strategy 1: "Superhero Brainstorm" Introduction.

- How to Implement: Begin by introducing the concept of "Turning Trash into Power" as a superhero. Ask students: "Every superhero needs villains to fight! If 'Turning Trash into Power' is our superhero, what two big 'villains' (problems in our world) do you think it would fight and solve when it comes to trash and energy?"
- Benefit: This approach engages students creatively, makes abstract environmental problems more tangible, and directly addresses the first challenge question.

Strategy 2: Think-Pair-Share: The Energy Link.

- How to Implement: Ask students: "When waste burns and gets super-hot in special power factories, what does that heat mostly make that we can use as energy?" Give them time to think individually, then pair them to discuss their ideas, and finally share answers with the whole class.
- Benefit: Encourages individual reflection and collaborative discussion, leading to a deeper understanding of the core process of waste-to-energy, directly addressing the second challenge question.

Strategy 3: Guided Discussion Using Visuals: The Power Plant Journey.

- How to Implement: Show a simple diagram or animation of a waste-to-energy plant. Point to each step (e.g., waste arriving, burning, heat creating steam, steam turning a turbine, generator making electricity). Verbally guide them through the process: "This is where the trash comes in... then it gets very hot... and that heat makes steam! What does the steam do next?"
- Benefit: Provides a visual aid for a complex process, helping students grasp how heat is converted into usable energy.

Strategy 4: "Cause and Effect" Sentence Starters

- How to Implement: Provide sentence starters to help students articulate their understanding:
 - "Turning trash into power helps fight pollution by..."
 - "When trash gets hot, it makes steam, which then..."
 - "One problem waste-to-power solves is..."
- Benefit: Supports students in structuring their thoughts and using appropriate vocabulary when explaining the concepts.



Builder

Objective:

Students will design and construct a circuit where an LED lights up to clearly show when the trash bin is full, demonstrating their understanding of how a sensor can trigger a visual indicator.

- This supports creativity, problem-solving, and the real-world application of basic electronics principles related to sensors, outputs, and automated alerts.

Materials Needed:

- Arduino Uno Board.
- Breadboard
- IR Infrared Sensor Module (to detect fullness).
- LED (Light Emitting Diode).
- Resistor (220-330 ohm, for LED protection).
- Jumper Wires (Assorted lengths).

Activity Instructions:

1. Review the Goal:

- Remind students they are building a crucial part of their smart bin: a circuit where an LED light will turn on to clearly show when the trash is full.

2. Plan Your Design:

- Ask students to look at a provided circuit diagram (like the example showing an IR sensor and LED connected to Arduino).
- Discuss: How will the IR sensor "see" the trash? How will the LED turn on when the trash is detected?

3. Gather Materials:

- Provide the Arduino, breadboard, IR sensor, LED, resistor, and jumper wires.
- Encourage students to identify each component and its role.

4. Build the Circuit:

- Safety First: Remind students to handle components carefully and ensure the Arduino is unplugged before making connections.
- Connect Power (Arduino to Breadboard):
 - Connect the 5V pin from the Arduino to the positive (+) rail on the breadboard.
 - Connect a GND pin from the Arduino to the negative (-) rail on the breadboard.
- Connect the IR Sensor:

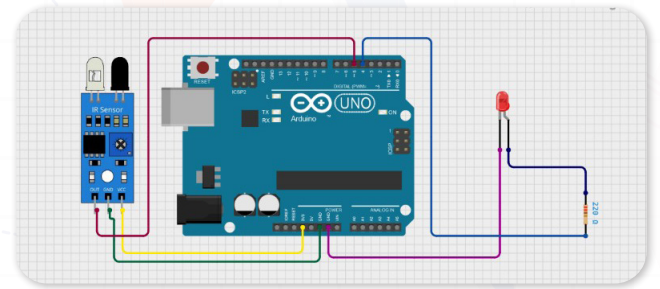


Figure 30: Arduino connected to an IR sensor and an LED.

- Connect the IR sensor's VCC pin to the positive (+) rail on the breadboard.
- Connect the IR sensor's GND pin to the negative (-) rail on the breadboard.
- Connect the IR sensor's OUT (signal) pin to a digital pin on the Arduino (Digital Pin 7).
- Connect the LED Indicator:
 - Place the LED on the breadboard (long leg positive, short leg negative).
 - Connect one end of the resistor to the short (negative) leg of the LED.
 - Connect the other end of the resistor to the negative (-) ground rail.
 - Connect the long (positive) leg of the LED to another digital pin on the Arduino (Digital Pin 2).

Possible Strategies for Implementation:

1. Inspiration First: Real-World Indicators:

- How to Implement: Show photos or short videos of real-world "full" indicators (e.g., a "paper full" light on a printer, a "full tank" light on a car dashboard, a battery low indicator).
- Benefit: Helps students connect their circuit to practical applications they already understand, showing the purpose of a visual alert.

2. Peer Collaboration: Circuit Building Pairs:

- How to Implement: Pair students or use small teams to work together on assembling the circuit. One student can read the instructions while the other connects components, then switch roles.
- Benefit: Encourages teamwork, shared problem-solving, and allows students to learn from each other during the hands-on building process.

3. Guided Questions: Connecting Cause and Effect:

- How to Implement: Throughout the building and testing, ask questions like:
 - "What does the IR sensor 'see' when the trash is full?"
 - "How does that 'seeing' make the LED turn on?"
 - "Why is it important for the LED to be bright and clear?"
 - "What did you learn about how electricity makes the LED turn on when the sensor detects something?"
- Benefit: Prompts students to explain their thinking, reinforce cause-and-effect relationships in circuits, and solidify their understanding of the system.

4. Visual Checklists & Diagrams:

- How to Implement: Provide clear, step-by-step visual diagrams of the circuit connections. Use a checklist for each component to ensure correct placement.
- Benefit: Offers multiple means of representation, supporting visual learners and providing structure for complex tasks.

5. Safety Check Prompt:

- How to Implement: Before students plug in their Arduino, have them perform a "Safety Check." Prompt: "Are all wires firmly connected? Is the resistor in place with the LED? Is anything touching that shouldn't be?"
- Benefit: Reinforces safe practices when working with electronics.



Mastermind

Objective:

Students will apply their understanding of programming logic and electronic components by arranging scrambled lines of code to control their smart bin's full-indicator LED. This activity will help them practice critical thinking, develop coding skills, and troubleshoot to bring their electronic project to life.

Answers:

```
int ledPin = 7;
int sensorPin = 2;
int sensorValue = 0;
void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(sensorPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  sensorValue = digitalRead(sensorPin);
  if (sensorValue == HIGH) {
    digitalWrite(ledPin, HIGH);
    Serial.println("Bin is FULL!");
  } else {
    digitalWrite(ledPin, LOW);
  }
  delay(500); // Optional: helps avoid rapid flickering or message spam
}
```

Activity Instructions:

1. **Introduce the Mission:** "You've learned about the amazing 'brains' (like Arduino) and 'muscles' (like Servos) of smart projects, and even how our 'eyes' (the IR sensor) work! You've even built the circuit! Now, it's time to translate your brilliant design into action! Your mission is to arrange the lines of code to bring your circuit's intelligence to life!"
2. **Clarify the Goal:** "When you finish arranging and uploading your code, here's what should happen: the circuit will turn on the LED when the bin is full (simulated by the sensor)."
3. **The Task:** "Now it's time to turn your smart design into real action! Cut out the lines of code, scramble them, and let students arrange them in the correct order to complete the program." (Referencing code from image_a2f73b.png)
4. **Emphasize Logic:** "This isn't about guessing! It's like writing a recipe for our Arduino. Think about what needs to happen first, what happens next, and what happens over and over again for our smart system to work. Every line of code tells the Arduino exactly what to do!"

Possible Strategies for Implementation:

Strategy 1: "Code Blueprint: Outline First!"

- **How to Implement:** Before arranging the code, have students write a simple "blueprint" or flowchart of their code's logic on paper. For example:
 - Setup: Tell Arduino about LED pin, sensor pin. Set pin modes (input/output). Start Serial communication.
 - Loop: Read sensor value.
 - Decision: IF sensor value indicates "full" (e.g., HIGH), THEN turn LED ON and print "Bin is Full!".
 - ELSE (if not full), THEN turn LED OFF.
- **Benefit:** This helps students structure their thoughts logically before diving into syntax, mimicking real-world software development.

Strategy 2: "Paired Programming: Two Heads, One Code!"

- **How to Implement:** Have students work in pairs. One student acts as the "driver" (physically arranging the code snippets) and the other as the "navigator" (guiding the driver, reviewing lines, spotting logical errors). Encourage constant communication.
- **Benefit:** Fosters collaboration, allows for immediate error checking, and promotes shared learning.

Strategy 3: "Test as You Go: Conceptual Testing!"

- **How to Implement:** Encourage students to arrange a small part of the code (e.g., just the variable declarations and setup()), and conceptually "run" it in their minds or discuss its purpose. Then add the loop() structure, then the if/else logic.
- **Benefit:** Breaks down the coding task into manageable chunks, making logical troubleshooting easier and building confidence with each successful step.

Strategy 4: "Debugging Detectives: What Went Wrong?"

- How to Implement: When code doesn't seem to make sense in their arrangement, guide students through "debugging" their logic. Ask: "What did you expect to happen with these lines? What actually happens if we put them in this order? What line of code controls that specific action?"
- Benefit: Develops essential troubleshooting and analytical skills, turning errors in arrangement into learning opportunities.

Strategy 5: "Code Storyteller: Read It Aloud!"

- How to Implement: Once students have arranged their code, have them "read" it aloud line by line, explaining in simple terms what each line tells the Arduino to do and how it achieves the "full" indicator goal.
- Benefit: Reinforces their understanding of syntax and logic, helping them "predict" the circuit's behavior from the code.

Elaborate



Teacher Role:

- Facilitator: Guide students through the process of carefully placing and integrating their previously built smart bin circuit (IR sensor, LED, and servo motor) into their physical bin design. Organize a "Smart Bin Demo Day" where students showcase their functional bins.
- Observer: Observe how students safely and effectively integrate the electronic components into their physical bin models, noting the functionality of the IR sensor, LED indicator, and servo motor (for lid action, if applicable).
- Questioner: Ask students to explain:
 1. "How did you secure each component (sensor, LED, servo, Arduino) inside the bin?"
 2. "Why did you choose that specific placement for the IR sensor to detect fullness?"
 3. "How does the LED clearly show when the bin is full?"
 4. "How does the servo motor (if used) make the bin 'smart'?"
 5. "What challenges did you face when putting the electronics inside your bin, and how did you solve them?"
- Feedback Provider: Offer constructive feedback based on a rubric, focusing on the neatness, robustness, and functionality of the integrated circuit within the bin, and its effectiveness as a "smart" waste solution.
- Connector: Help students connect their final integrated project back to all previous lessons: understanding sensor input, programming logic, visual indicators, mechanical outputs, and the overall problem-solving process for a real-world application.

Expected From Students:

- **Circuit Integration:** Students will demonstrate the successful and secure integration of the IR sensor, LED, and servo (if used) circuit within their physical smart bin design.
- **Secure Placement:** Students will carefully place and mount all electronic components (Arduino, breadboard, sensors, output devices) within their bin, ensuring wires are tidy and connections are stable.
- **Explanation of Functionality:** Students will be able to explain how the entire integrated system works: how the IR sensor detects fullness, how the LED indicates it, and how the servo (if used) acts as an automated lid or dispenser.
- **Problem-Solving Application:** Students will apply problem-solving skills to overcome challenges related to component placement, wire management, and ensuring reliable operation within the physical bin structure.
- **Demonstration:** Students will be able to demonstrate their functional smart bin, showing how it responds to "trash" (simulated by an object) and indicates its full status.

Parts of the books included.

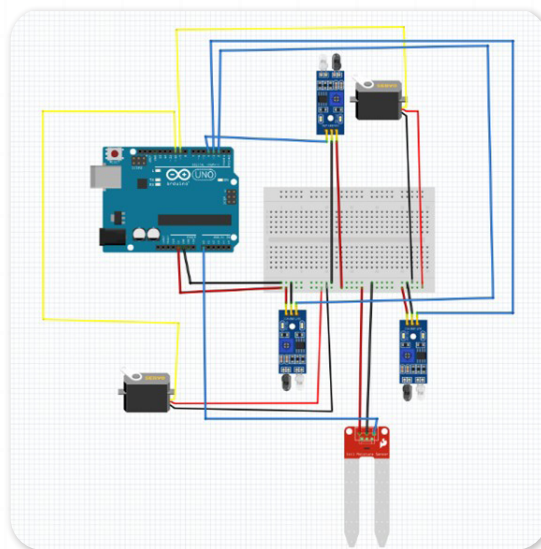


Figure 31: Complete Smart Trash Bin Circuit.



Showcase

Objective:

Students will apply their knowledge of the smart bin's components and their functions by engaging in a game that demonstrates the fully integrated circuit (featuring the IR sensor, LED indicator, and potentially a servo motor). This activity encourages interactive observation, understanding the practical application of smart technology, and reinforcing how the bin's "intelligence" works in a tangible, playful way.

Activity link: <https://g.co/gemini/share/ce81e4b180e1>



Note: (the students will not connect the whole circuit, you will describe the code to them and let them play the game).

Materials Needed:

- Teacher-Integrated Smart Bin Circuit:
 - Arduino Uno Board (with pre-uploaded code for smart bin functionality: IR sensor, LED, optional servo).
 - IR Infrared Sensor Module.
 - LED (with resistor).
 - Servo Motor (if integrated for lid automation).
 - Breadboard (with all components securely connected).
 - Jumper Wires (neatly managed).
 - USB Cable (for connecting Arduino to demonstration computer).
 - Power supply for Arduino (e.g., USB connection to computer, or battery pack if demonstrating wirelessly).
- For the Coding Game:
 - Computers, laptops (one per student or pair).
 - Internet access.
 - (Optional) Projector or large screen to display the game for whole-class instruction/ demonstration.

The code:

```
#include <Servo.h>
// IR Sensors (detect objects and fullness)
const int irSensor1 = 2; // Detects object near bin
const int irSensor2 = 3; // Detects fullness of wet waste part
const int irSensor3 = 4; // Detects fullness of dry waste part

// Soil Moisture Sensor
const int soilSensor = A0;
// Servo Motors
Servo lidServo; // Controls bin lid
Servo sortServo; // Controls waste sorting

//LED for Alerts
const int fullBinLED = 6;
```

```
// Moisture Threshold (adjust based on testing)
const int moistureThreshold = 400;

void setup() {

  // Set IR sensor pins as input
  pinMode(irSensor1, INPUT);
  pinMode(irSensor2, INPUT);
  pinMode(irSensor3, INPUT);

  // Set LED pins as output (these were missing in original code)
  pinMode(fullBinLED, OUTPUT);

  // Attach servo motors
  lidServo.attach(9);
  sortServo.attach(10);

  // Initial positions
  lidServo.write(0); // Lid closed
  sortServo.write(90); // Neutral sorting position

  // Ensure LED is off initially
  digitalWrite(fullBinLED, LOW);
}

void loop() {
  // Read IR sensor values
  int trashDetected = digitalRead(irSensor1); // Object detected near bin
  int wetBinFull = digitalRead(irSensor2); // Wet waste bin full check
  int dryBinFull = digitalRead(irSensor3); // Dry waste bin full check

  // Read soil moisture sensor
  int moistureValue = analogRead(soilSensor);

  // Display sensor readings in Serial Monitor
```



```

// Display sensor readings in Serial Monitor
Serial.print("Moisture: ");
Serial.print(moistureValue);
Serial.print(" | Trash Detected: ");
Serial.print(trashDetected);
Serial.print(" | Wet Bin Full: ");
Serial.print(wetBinFull);
Serial.print(" | Dry Bin Full: ");
Serial.println(dryBinFull);

// If any bin is full, trigger alert
if (wetBinFull == HIGH || dryBinFull == HIGH) {
  Serial.println("One of the bins is full! Empty the bin.");
  lidServo.write(0); // Keep lid closed
  digitalWrite(fullBinLED, HIGH); // Turn on full bin LED
} else {
  digitalWrite(fullBinLED, LOW); // Ensure full bin LED is off if not full
}

// If trash is detected and bins are not full, open lid (AND only if full alert is not active)
// Added condition: !(wetBinFull == HIGH || dryBinFull == HIGH) to prevent opening if full
if (trashDetected == HIGH && !(wetBinFull == HIGH || dryBinFull == HIGH)) {
  lidServo.write(90); // Open lid
  delay(2000); // Wait for user to drop trash
  lidServo.write(0); // Close lid
}

// Sorting Logic: Wet vs. Dry Waste (only if trash is detected and not full)
if (trashDetected == HIGH && !(wetBinFull == HIGH || dryBinFull == HIGH)) {
  if (moistureValue > moistureThreshold) { // Wet Waste
    Serial.println("Sorting: Wet Waste");
    sortServo.write(180); // Move to wet waste section
  } else { // Dry Waste

```

```
Serial.println("Sorting: Dry Waste");
sortServo.write(0); // Move to dry waste section
}
delay(2000); // Wait before resetting to neutral position
sortServo.write(90); // Reset position
}

delay(500); // Small delay for sensor readings before the loop repeats
}
```

Activity Instructions for Teachers:

1. Introduce the Smart Bin & Coding Game:

- "Today, we're going to understand the 'brain' of our smart bin even better by playing a special coding game! I've already integrated the circuit inside a demonstration setup, so we can focus on how the code makes it smart."
- Briefly demonstrate the pre-integrated smart bin circuit's functionality (e.g., show how placing a hand near the sensor triggers the LED to light up, and the servo to move).

2. Explain the Game's Purpose & Rules:

- Introduce the coding game students will be playing.
- Clearly explain the game's objective (e.g., arranging code blocks, writing lines of code, or making choices that mimic the smart bin's logic).
- Go over the rules, scoring (if any), and how success in the game relates to understanding the smart bin's automation.

3. Guide Game Play:

- Lead students to the computers/tablets with the coding game loaded.
- Provide initial instructions on how to navigate the game interface and manipulate code.
- Circulate around the room, offering support and answering questions as students play.

4. Connect Game Logic to Real Circuit:

- As students' progress or after specific game levels, pause and ask guiding questions:
 - "How does this challenge in the game (e.g., making the light turn on) relate to our real smart bin's LED?"
 - "When the game asks you to make a decision 'if' the sensor sees something, how does that mimic what our IR sensor does?"
 - "Can you find the part of the game's code that would make the servo move, just like our smart bin's lid?"
- Refer back to the demonstration smart bin circuit as needed to clarify concepts.

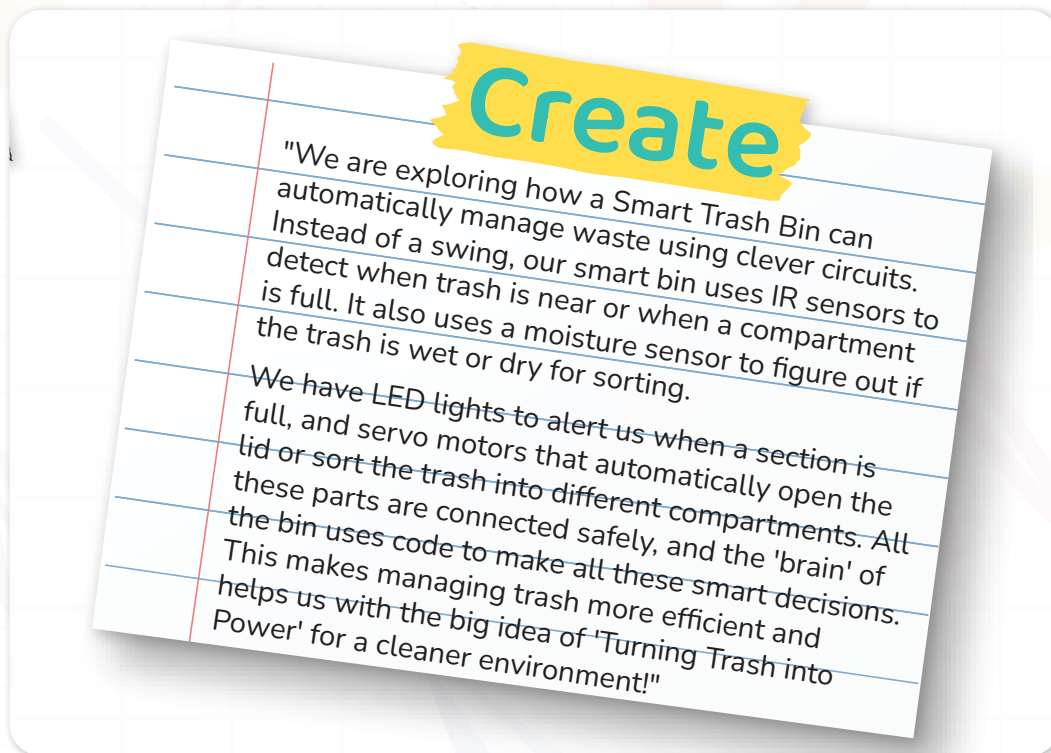
5. Facilitate Post-Game Reflection:

- Lead a discussion after students have had a chance to play the game.
- Ask: "What did this coding game teach you about how our smart bin knows when it's full and what to do next?"
- "How does writing code help make devices like our smart bin 'smart'?"
- "What was the most challenging part of the game, and what did you learn from it?"

Showcase: Rubric for evaluating the smart trash bin.

Criteria	Excellent (4 Point)	Good (3 Points)	Satisfactory (2 Points)	Needs Improvement (1 Point)
Circuit Assembly	Correct wiring, no errors, all components functional.	Minor wiring issues, mostly functional.	Some errors, needs debugging.	Major wiring issues, not functional.
Code Implementation	Code is correct, optimized, and well-commented.	Code works but needs slight optimization.	Code runs with minor errors.	Code has multiple errors, does not run.
Lid and Fullness Control	Lid opens/closes properly, fullness alerts work.	Minor response delays.	Works partially.	Lid does not respond properly.
Presentation & Explanation	Clearly explains each component and logic.	Explains most parts, but lacks clarity in some areas.	Basic explanation with missing details.	Struggles to explain functionality.
Innovation & Problem Solving	Adds creative features beyond requirements.	Implements basic requirements successfully.	Minimal innovation, meets only basic needs.	Struggles to implement basic logic.

Don't forget to go back to the EDP page and add these to create part.



Now I Can

Objective:

Students will reflect on their learning by identifying how the smart bin's sensor system works, how it automates actions, and how this technology helps in "Turning Trash into Power." This activity reinforces understanding of STEM concepts, encourages communication, and promotes critical thinking about smart solutions.

Activity Instructions:

1. Introduce the Purpose:

- Begin by explaining that students will reflect on everything they've learned and seen about the smart bin.
- Say: "This checklist helps you think about what you've done and what you've learned about our smart bin. It's okay if you can't check everything this shows what you're still working on!"

2. Student Reflection Time:

- Allow students quiet time to read each item (from the list below) and check the boxes they feel confident about.
- Encourage them to recall the smart bin demonstration, think about the coding game, or talk quietly with a partner if they need help remembering.

3. Optional Pair Discussion:

- Have students turn to a partner and share one or two things they are proud of being able to understand or explain from the checklist.

4. Class Share-Out (Optional):

- Invite a few students to share one item they checked and explain what they understood or what happened during the demonstration/game that helped them check it.

5. Teacher Observation/Documentation:

- Use this time to circulate and observe which skills students have mastered and which need reinforcement.
- You may choose to collect the checklists as formative assessment.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **3-5-ETS1-1:** Define a design problem that can be solved by developing an object or a system and that includes specified criteria for success and constraints on materials, time, or cost.
 - (Students identify the problem of waste management and how a smart bin offers a solution.)
- **3-5-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.
 - (Students analyze the smart bin's functionality based on demonstration and the coding game, understanding how different code logics lead to different automated behaviors.)
- **4-PS3-2:** Make observations to provide evidence that energy can be transferred from place to place by sound, light, heat, and electric currents.
 - (Students observe the transfer of electrical energy powering the sensors, LEDs, buzzer, and servo motors in the smart bin circuit.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening & Informational Text):

- **SL.4.1:** Engage effectively in a range of collaborative discussions (one-on-one, in groups, and teacher-led) with diverse partners on Grade 4 topics and texts, building on others' ideas and expressing their own clearly.
 - (Students will discuss their observations of the smart bin and their strategies during the coding game.)
- **RI.4.3:** Explain events, procedures, ideas, or concepts in a historical, scientific, or technical text, including what happened and why, based on specific information in the text.
 - (Students will explain how the smart bin operates, how its components function, and how the code dictates its procedures, based on demonstrations and the coding game.)
- **SL.4.4:** Report on a topic or text, tell a story, or recount an experience with appropriate facts and relevant descriptive details, speak clearly at an understandable pace.

- (Students will present and explain their understanding of the smart bin's automated functions and the logic from the coding game during showcase or share-out.)

3. Common Core State Standards (CCSS) - Mathematics (Mathematical Practices):

- **CCSS.MATH.PRACTICE.MP1:** Make sense of problems and persevere in solving them.
 - (Students apply problem-solving skills when engaging with the coding game challenges to make the virtual smart bin function correctly.)
- **CCSS.MATH.PRACTICE.MP4:** Model with mathematics.
 - (Students use the logic within the coding game to model the behavior of the real-world smart bin system.)